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Skeleton Based Abnormal Behavior Recognition Using Spatio-Temporal Convolution and Attention-Based LSTM

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Abstract

Behavior recognition is a well-known computer vision technology, it has been used in many applications such as video surveillance, robotics, human-computer interaction and sports video, etc. However, most of existing works ignored the depth information so that they resulted in over-fitting and the inferior performance. Consequently, a novel framework for behavior recognition is proposed which is based on spatio-temporal convolution and attention-based LSTM (ST-CNN & ATT-LSTM). In this framework, deep spatial information is merged into each segment, meanwhile we focus on the method of key information extraction, which is essential for improving behavior recognition performance. Finally, the proposed framework is evaluated with real world surveillance video data, and the results indicate that our framework is superior to existing methods.

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Keywords: Abnormal behavior; Attention; LSTM

1. Introduction

Behavior recognition has gain increasing interest in many research areas. It has been widely applied in public security area, including automatic identification, video surveillance, early warning system, etc. The purpose of behavior recognition is to analyze the behavior of human body correctly by the spatial and temporal characteristics extracted from the original video clips.

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Human behaviors involve many uncertain factors and the key issue is how to extract information features precisely. Behavior recognition algorithms can be divided into traditional algorithms and deep-learning based algorithms. Traditional behavior recognition algorithms usually extract features by artificial designed feature patterns and uses the extracted features as the input of the classifier to obtain behavior categories. The latest research trend in behavior recognition is based on a deep learning approach that automatically extracts features from raw visual data by using Convolutional Neural Networks (CNN). However, the parameters and weights of convolutional kernel need to be well trained, which determines the validity of the extracted feature.

Existing behavior recognition algorithms ignore the depth information and the importance of key actions in behavior recognition leading to time consuming and low accuracy. In details, spatial and temporal information [1] is the two key factors in behavior recognition, However, most of the behavior recognition research [2][3][4][5] using raw video RGB images which will lose spatio-temporal information in 3D space and it is easy to be affected by illumination, scale and occlusion. In some studies [6][7][8], many approximate methods are usually used to minimize additional damage caused by data problems.

Therefore, Since the human skeleton information is robust and not affected by illumination, scaling and occlusion in the video, we propose a behavior recognition method based on human skeleton information. The method extracts the depth information of the skeleton as the input data of the behavior recognition and minimize the additional damage of illumination, scale and occlusion. Then our model which is based on spatio-temporal relationship is applied to perform behavior recognition. The proposed method combines spatial information from a single skeleton with temporal information from multiple skeletons composing multiple temporal and spatial dimensions to improve behavioral recognition accuracy.

The rest of this paper is organized as follows: Section 2 describes current research of behavior recognition. Section 3 provides details of our proposed behavior recognition model. Section 4 evaluates our proposed framework with real world surveillance video data. Section 5 gives the conclusion of this paper.

2. Related work

Researches [9][10] show that the movement of a character can be represented by several highlights that describe the movement of the main joint. The behavior can be classified by the movement of key human nodes. The algorithms of behavior recognition through skeleton information are mainly divided into two categories: one is behavior recognition using manual extraction of features, the other is behavior recognition based on deep learning algorithms. However, the method of manually extracting features [11][12] relies on the prior knowledge of the designer and manual adjustment of algorithm parameters. Therefore, the method of deep learning is widely used in the behavior recognition algorithm based on skeleton sequence.

In recent years, CNN [13][14][15] has been proven to be effective in representing and recognizing human behaviors. Kim [16] designed a new Time-Converging Neural Network (TCN) based on CNN which can be used to identify 3D human behavior. Ke et al. [17] combined the new skeletal sequence representation with the CNN structure to identify behavior. Similarly, Wang [18] et al. proposed deep dual-flow ConvNet for behavior recognition. Yan [19] designed a graph-based CNN spatio-temporal convolution network (ST-GCN), The model uses graph theory to encode dynamic skeletons, but lacks processing of timing information. Li et al. [20] proposed to use a hierarchical CNN network to learn the spatial information between human joint points and dynamic features.

CNN has good performance on images [21][22], but it is still difficult to remember the entire sequence [23] containing a lot of time step information and it is difficult to construct a deep network to extract advanced features. Considering that recurrent neural networks (RNN) [24][25] can simulate long-term context information [26] of time series well, Du et al. [27] proposed a skeleton-based behavior recognition network (HBRNN). Condering the LSTM [28][29] is better able to learn long dependencies. Yang [30] proposed a MTL mechanism that combines 3D CNN and LSTM to adapt to the information sharing of video input. Xue et al. [31] proposed a view adaptive RNN that uses two LSTM sub-networks to regress the spatial rotation parameters and spatial translation parameters of the skeleton, the parameters were sent to the primary LSTM network to predict behavior category.

3. ST-CNN & ATT-LSTM

In this section, a skeleton behavior recognition model based on spatio-temporal relationships is proposed. The model includes a target depth estimation algorithm for the skeleton and a network combining spatio-temporal convolution and attention-based LSTM (ST-CNN & ATT-LSTM). The model structure is shown in Figure 1.

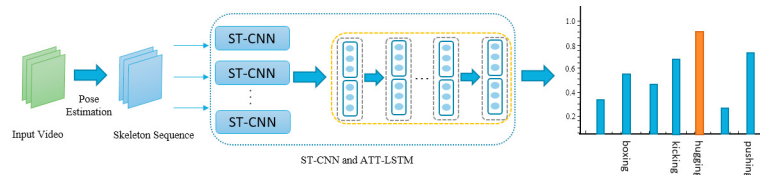


Fig. 1. We perform pose estimation from the input video and obtain a skeleton sequence. The skeleton sequence and its depth information are inputs to ST-CNN and ATT-LSTM. Finally, it will be classified into the corresponding behavior category.

3.1. Target depth estimation algorithm based on monocular fixed camera

In order to realize the mapping of depth measurement and 2D skeleton information to 3D skeleton information, a new method is proposed. This method includes two steps to extract skeleton depth information. They are: i) Initial value acquisition. ii) Calculate the distance between each joint point of the target and the camera. Since the overhead angle is not deformed when the camera is photographed under the fixed camera, the angle between the camera to the target foot and the camera to the target can be calculated according to the height of the person in the shooting screen.

3.1.1. Initial value acquisition

The camera is mounted in the corner of the room as shown in Figure 2 (a). The spatial view of the 3D scene is shown in Figure 2 (b).

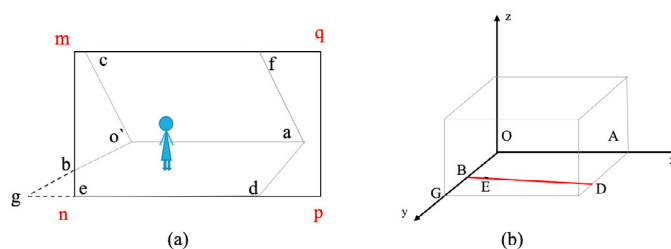


Fig. 2. 2D and 3D map

The lower left corner of the shooting picture is the origin of the 2D coordinate system. We set (x_{ii}^l, y_{ii}^l) as the coordinates of the target foot point l_i^l and (x_{ii}^h, y_{ii}^h) as the coordinates of the target head h_i^l point in the 2D image. Pentagonal $ao'bgd$ in Figure 2 (a) is the ground range of the shooting picture. In the real world, the actual distances of $o'-a$, $o'-g$, $a-d$, and $g-d$ can be measured manually.

3.1.2. Calculate the distance between each joint point of the target and the camera

In a 2D image, the distance between each pixel on the ground and the camera is uniquely determined. Therefore, the coordinates of the target foot can be used to calculate the distance between the target and the camera. We use the formula to calculate the distance between the target's foot and the camera. The formula is $x'_{ii} = f_1(x'_{ij})$ and

$y'_{ii} = f_2(y'_{ij})$, where f is a mapping function for shooting ground coordinates and real ground coordinates. Given the coordinates of four pairs of pixels corresponding to projection transformation, the projection transformation matrix can be calculated.

We calculate the angle between the camera and each joint point of the target based on the height of the person in the captured picture. According to the sine theorem of the triangle, the distance between the joint point and the camera can be calculated. The positional relationship between the person and the camera is as shown in Fig. 3, where a is the position of the camera, f and g are the real positions of the target head and the foot, and d and e are the positions of the target head and the foot in the capture picture. It can be seen from Figure 3 that the angle between the target $f-g$ and the camera is the same with the angle between the target $d-e$ and the camera in the projection screen, and they both are angles α .

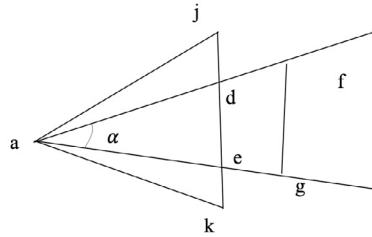


Fig. 3. stadia map of camera

For the determined target, the coordinates of the j and k points in the 2D picture can be accurately calculated from the intersection of the two lines, and i represents the target in the captured picture. The 3D coordinates of the point j and the point k in the real scene can be obtained by mapping the size ratios of mn , np , pq , and pq in the captured image to the 3D space. The coordinate position of the camera point a in the 3D space is known by the previous measurement, and the distance d_{aj} from the camera to the upper boundary point of the target line and d_{ak} from the camera to the lower boundary point of the target line can be calculated in figure 3. The three apex angles of the triangle $\angle jak$, $\angle akj$, $\angle ajk$ can also be obtained by the cosine formulas.

As shown in Figure 3, for the determined triangle $\angle akj$, the distance ratio of jd , de and ek is known, and the true distance of jd , de , ek can be obtained. The geometrical relationship can be used to find the angle $\angle dae$ between the camera to the target and the camera to the target which is the angle α . The calculation formula is as follows:

$$\alpha = \arccos\left(\frac{d_{aj}^2 + d_{ak}^2 - d_{jk}^2}{2d_{aj} \times d_{ak}}\right) - \arcsin\left(\frac{d_{jd} \times \sin \angle ajk}{d_{ad}}\right) - \arcsin\left(\frac{d_{ek} \times \sin \angle akj}{d_{ae}}\right) \quad (1)$$

By calculating the real distance from the target foot point to the camera and the line of sight angle α from the camera to the target foot and the camera-to-target, the true height H of the target can be solved by the geometric relationship.

3.2. Skeleton behavior recognition model based on spatio-temporal relationship

In this section, we propose a skeleton behavior recognition model based on spatio-temporal relationships, including spatio-temporal convolution (ST-CNN) and attention-based LSTM (ATT-LSTM). It combines the characteristics of CNN and LSTM as well as the spatially related properties of the skeleton sequence itself. The proposed network framework is shown in Figure 4.

3.2.1. Spatio-temporal convolutional neural network based on self-attention

In the ST-CNN part, the spatio-temporal convolution operation is performed based on the input data. The skeleton sequence is split up into n consecutive groups which are the input of the network. We set the input sequence as $X = (x_1, x_2, \dots, x_n)$, n represents the number of continuous skeleton subsequence, x_i represents joint data of human

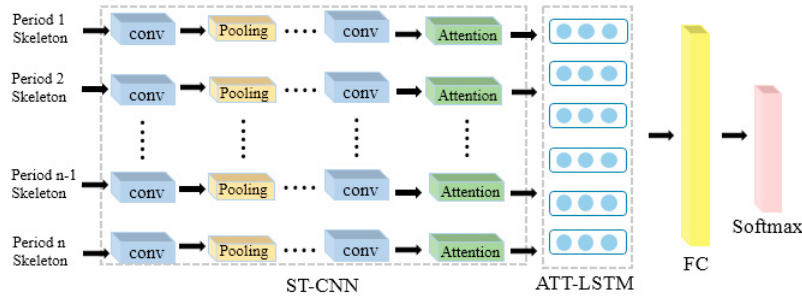


Fig. 4. Skeleton behavior recognition network model diagram based on spatio-temporal relationship

body during the i -th phase which is a $D \times M \times T$ dimensional matrix. D is the dimension of the human joint data, since the input is 3D skeleton data, we set D to 3. M denotes the number of skeleton joints and T represents the number of frames of the skeleton sequence. The filter dimension is $3 \times 3 \times t$, and t is the time depth of the kernel. The size of the first pooling layer is set to $3 \times 3 \times 1$, and the rest is $3 \times 3 \times 3$. The last number is the time depth, and setting it to 1 means pooling on a single frame, which will be beneficial to retain the time feature. Since the convolution operation is performed on both the time and spatial dimension, the local spatio-temporal feature information of the skeleton sequence can be obtained in a short time.

After a small group of the skeleton sequence has undergone the above convolution operation and pooling operation, a spatio-temporal feature map of n time periods is acquired. These feature maps indicate a set of spatio-temporal features in the $\frac{1}{n}$ time period. The dimension of this feature map is $N \times M \times K$. The K above is the feature dimension of every joint. We get the weighted features of these joints by using self-attention which can focus the model on more important joints. Each phase contains a self-attention layer followed by a feedforward neural network:

$$f_n^{(1)} = \left(FNN(MultiHead(f_n^{(0)}, f_n^{(0)}, f_n^{(0)})) \right) \quad (2)$$

FNN is the feedforward neural networks and the $f_n^{(1)}$ is the weighted features of the skeleton sequence.

3.2.2. Attention-based LSTM

In order to deal with the spatio-temporal relationship, we propose a attention-based LSTM algorithm. ATT-LSTM allows the model to focus more on important frames. The network structure of ATT-LSTM is shown in Figure 5.

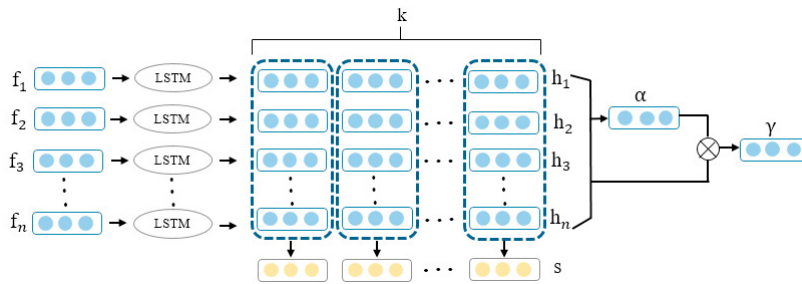


Fig. 5. ATT-LSTM network diagram

In the process of behavior recognition, we pay different attention to each frame of the video. For example, when faced with fighting behavior, the target pays more attention when punching and kicking appear. In this sequence, the behavior of two people before and after kicking is relatively small in the determination of the skeleton sequence. The proposed ATT-LSTM can deal with scenes that require different attentions in different time periods. The spatio-temporal features of each successive time period obtained by ST-CNN are used as input of ATT-LSTM to obtain the spatio-temporal features of the global skeleton sequence.

N feature sets $f = [f_1, f_2, \dots, f_n]$ are obtained by local spatio-temporal convolving neural networks, and the N features is used as the input of the ATT-LSTM network. Each time ATT-LSTM receives a spatio-temporal feature map, the first half of the ATT-LSTM is the LSTM network, and a high-level feature map is available for each moment of input. Combine all the time feature maps output by the LSTM, which can be expressed by formulas (2), (3), (4) and (5):

$$h_i = g(f_i), i \in (1, n) \quad (3)$$

$$Q = [h_1, h_2, \dots, h_n] \quad (4)$$

$$\alpha = \text{softmax}(w^T \tanh(Q)) \quad (5)$$

$$r = Q\alpha^T \quad (6)$$

Where the g function in equation (3) represents the feature extraction process of LSTM, and equation (4) is the result of the combined output of all time feature maps output by the LSTM. Using the local features of the time series as input to the LSTM, the LSTM forgetting gate means that the useless segment in the previous time period is removed, and the input gate increases the useful segment in the current time segment. The LSTM gate represents the selection process in the skeleton sequence, retains useful time period features, and discards useless time period features so as to obtain the most useful timing features in the skeleton sequence. The α in equation (5) represents the attention of the n features in the output of the LSTM, and the equation (6) represents the weighted summation of the attention at the LSTM output, in this equation the more important features give higher attention.

Due to the continuous activation function processing, the useful information obtained will increase over time, so that the information retained in the output h_n at the last moment is less and less. In other words, the closer the output h_n at the last moment, the closer the time is, the more useful information is retained. Therefore, in order for the network to get all the useful information on average, the ATT-LSTM design averages all the output to get the global feature map as the second half of the network. The formula is as shown in equation (7):

$$S_j = \frac{\sum_{i=1}^n h_{ij}}{n}, i \in (1, n), j \in (1, k) \quad (7)$$

In equation (7), S is the global feature representation of the entire skeleton sequence. Since the input into the LSTM network is a local spatio-temporal feature map of the skeleton extracted by the convolutional network. Therefore, while acquiring the spatio-temporal features of all skeletons, the calculation amount of LSTM is greatly reduced. Compared with directly training the entire skeleton sequence into the LSTM network for training, the efficiency of network training and operation is improved.

The third part of the network is connected to the fully connected layer with softmax. The size of the fully connected layer is the size of the category to be classified, and the global feature map is input into the part to obtain the result of the final behavior classification.

4. Experiment

4.1. Dataset

In order to evaluate the effectiveness of the proposed method, we compare the proposed model with other models in the SBU interaction dataset and the proposed LF-skeleton+3DInfo dataset. The LF-skeleton+3DInfo dataset is a 2D skeleton information plus skeleton depth information, in which the 2D skeleton data is extracted by Openpose [32] method on LF-skeleton, and the depth estimation method is used to obtain the 3D skeleton dataset of depth information.

SBU Interaction Dataset [33]: The same lab environment in which two people interact to capture data sets. The dataset consists of approximately 300 sequences of 8 action types, each consisting of 15 joints. The data set consists of 21 collections, each consisting of a pair of different people who perform all eight operations.

LF-Skeleton+3D info Dataset: Since the scene developed in this paper is undisclosed, it is incapable to obtain the behavior dataset under the real prison scene. In addition, most of the currently disclosed skeleton-based behavior

datasets are based on single-person, and the shooting angle of the camera does not meet the requirements of the actual multi-person scene. Therefore, in order to make better use of deep learning to train a model that conforms to the scene of the institution, this paper designs and produces a multi-person interaction behavior dataset that simulates the scene of the prison. The dataset has a total of 400 sample sequences, a total of 8 types of interaction. The dataset was captured by 10 participants in a fixed camera scene, each of which was done by different participants, each with double and multiplayer interactions. Among them, there are about 320 sample sequences for double interaction and 80 for multi-person interaction. Each interaction lasts for about 3 seconds. In order to obtain the skeleton dataset, the 2D skeleton is acquired by the real-time multi-person pose estimation algorithm (OpenPose) based on the affinity field proposed by Carnegie Mellon Robot Research Center [32]. Each sample sequence contains a skeleton sequence of about 50 to 80 frames, and the acquired skeleton data set is data-corrected to obtain a more complete and effective skeleton dataset.

4.2. Experiment analysis

In addition to the identification of the dataset Top1, this paper also divides the LF-Skeleton+3DInfo dataset into two cases: abnormal behavior and non-abnormal behavior. The dataset includes eight interactions of pushing, boxing, kicking, hugging, approaching, moving away, shaking hands, and exchanging items. The three behaviors of kicking, pushing, and boxing in the dataset are defined as abnormal behaviors, and other behaviors are defined as Non-abnormal behavior, that is, normal behavior. Then we identify the abnormal behavior of the LF-Skeleton+3DInfo dataset after the binary classification. The training process of the model is still identified by the eight behavioral actions in the LF-Skeleton+3DInfo dataset. The result of the abnormal recognition rate is calculated according to whether the binary classifications are correct or not. The results are shown in Table 2.

Table 1. Experimental results of ST-CNN&ATT-LSTM on LF-skeleton and LF-skeleton 3D-info dataset.

method	Acc.(%)	3D-Info Acc.(%)
Temporal Conv [16]	41.1	45.3
ST-GCN [19]	46.3	50.1
Ours	50.5	54.7

Table 2. Experimental results of Top1 and anomaly classification of ST-CNN&ATT-LSTM on LF-Skeleton+3DInfo dataset.

method	Top-1(%)	AbnormalAcc.(%)
ST-CNN&ATT-LSTM	50.5	81.1
ST-CNN&ATT-LSTM+3D-Info	54.7	84.2

Table 1 shows the experimental results and comparison of ST-CNN & ATT-LSTM on the LF-skeleton dataset and LF-skeleton + 3D-info dataset. Table 2 shows the experimental results of top1 and anomaly classification of ST-CNN & ATT-LSTM on the LF-Skeleton+3DInfo dataset. The 3D information is an experimental result based on a target depth estimation algorithm of a monocular fixed camera. As can be seen from Tables 1 and 2, in the case of using the model proposed (ST-CNN & ATT-LSTM) in this paper, the accuracy of using the depth estimation method is increased by 4% compared to the accuracy of using the 2D skeleton as an input. It can be seen from Table 1 that in other behavior recognition models based on skeleton sequences, using the data containing depth information as input, the effect of the model is also improved. The experimental results show that the depth estimation method can improve the accuracy of skeleton behavior recognition, and prove that the depth estimation method is feasible.

Table 3 compares the proposed ST-CNN & ATT-LSTM model with the advanced methods in the SBU Interaction dataset in recent years. As can be seen from Table 3, the skeleton-based behavior recognition model based on spatio-temporal relationship (ST-CNN & ATT-LSTM) proposed in this paper is higher than the traditional skeleton behavior recognition method. At the same time, this method is also at the advanced level in the existing deep learning algo-

Table 3. Experimental results and comparison of ST-CNN & ATT-LSTM on SBU Interaction dataset.

method	Acc.(%)
Joint feature [33]	80.3
Hierarchical RNN [27]	80.35
Ours	83.3

rithms, which is higher than the accuracy of most deep learning methods. It shows that the proposed skeleton-based behavior recognition model based on spatio-temporal relationship (ST-CNN & ATT-LSTM) is feasible.

5. Conclusion

In this paper, we proposed a framework based on skeleton spatio-temporal relationship and attention-based behavior recognition (ST-CNN & ATT-LSTM). Based on the target depth estimation algorithm of the monocular camera, the framework acquires the depth information of the human body's 2D skeleton and identifies the behavior through the temporal and spatial relationship of the skeleton.

The fixed camera based depth estimation method simulates real indoor scene. Through the 2D image and the real indoor spatial information, the geometrical method is used to calculate the 3D spatial position information of the target. The method uses a small amount of data to quickly and accurately convert the 2D skeleton information to the 3D skeleton information and provides data preparation for improving the accuracy of algorithm analysis.

The spatio-temporal relationship-based skeleton behavior recognition model uses a convolutional neural network based on self-attention to obtain the weighted spatio-temporal features of the skeleton sequence. An improved LSTM method is proposed to obtain the spatio-temporal features of long-term skeleton sequences so as to better deal with long-term skeleton sequences. It can be seen from the experimental comparison that the accuracy of the algorithm is higher than that of the traditional model and is at an advanced level in the deep learning model.

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